

Chapter 9: Relativistic Field Theory:

Exercise #1:
(pg 527)

@ $t = 0$

$$X'^{\mu} = \Lambda^{\mu}_{\nu} X^{\nu}$$

(from problem) "Lorentz Boost"

$$\Lambda = e^{\theta K_1}$$

(pg 527) "Symmetric matrices"

$$K_2 = i(E^{01} + E^{10})$$

$$K_1 = i(E^{01} + E^{10})$$

$$= i \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E$$

$$X'^{\mu} = \Lambda^{\mu}_{\nu} X^{\nu}$$

$$= e^{\theta K_1} \cdot X^{\nu}$$

$$= e^{i\theta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E} \cdot X^{\nu}$$

$$= (1 + i\theta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E - \theta^2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E^2) X^{\nu}$$

$$= X^{\nu} + i\theta \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E \cdot X^{\nu} - \theta^2 \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} E^2 \cdot X^{\nu}$$

$$\cong X'^{\mu}$$

Common Lorentz

Boost in 1-dimension:

$$t' = \gamma \left(t - \frac{vX}{c^2} \right)$$

$$X' = \gamma (X - vt)$$

where γ = "Lorentz Factor"

$$= \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$$

t = "time"

v = "velocity"

c = "speed of light"

Exercise #2:
(pg 523)

(9.7) "Commutator Relations"

$$[J_i, J_j] = i \epsilon_{ijk} J_k$$

$$[K_i, K_j] = -i \epsilon_{ijk} J_k$$

$$[J_i, K_j] = i \epsilon_{ijk} K_k$$

(9.8) "S.O. Rotation and Boost"

$$J_i = -i \epsilon_{ijk} E^{jk}$$

$$K_i = i (E^{0i} + E^{i0})$$

Exercise #3: Hermitian Values:

(pg 531)

(9.26) " γ -matrices"

$$\gamma^0 = \tau_1$$

$$\gamma^i = -i\tau_2 \otimes \sigma_i$$

$$\bar{\gamma}^{i+} = i\tau_2 \otimes \sigma_i$$

$$\neq \gamma^i$$

Definition:

Hermitian - a complex square matrix that is equal to its own conjugate transpose.

Traceless values:

γ^μ is traceless because no diagonal elements in the Lorentz boost.

Exercise #4: (from problem) "Inner Product"

(pg 548)

$$\langle f, g \rangle = \int d^4x \bar{F}g = \int d^4x F^\dagger \gamma^0 g$$

(pg 548) "Dirac operator"

$$\not{D} \equiv (\not{\partial} - iA_\mu) \gamma^\mu$$

Hermitian calculation:

$$\begin{aligned} \langle f, (i\not{D})g \rangle &= \int d^4x F^\dagger \gamma^0 (i\not{D})g \\ &= \int d^4x F^\dagger \gamma^0 (i(\not{\partial} - iA_\mu) \gamma^\mu)g \\ &= \int d^4x (-i \gamma^\mu (i\partial_\mu + iA_\mu)) F^\dagger \gamma^0 g \end{aligned}$$

$$= \int d^d x \overline{(i\partial)F} g$$

$$= \langle (i\partial)F, g \rangle$$

Quantum Mechanics Postulate #2:

"Every observable corresponds to a real Hermitian operator"

Cite: Born, Max V.

"Zur Quantenmechanik II"

Göttingen, Germany, (1925)

Observables are real. Also, Hermitian determinants are real, along with eigenvalues. The absence of a real spectrum is not Hermitian.

The inner product above never excludes Hermitian operators with complex eigenvalues because the product by f and g . The eigenvalue measures real properties in a system from the inner product.

Exercise #5: (9.44) "Inner Product"

(pg 548)

$$\langle f, g \rangle = \int d^d x \bar{f} g = \int d^d x f^\dagger \gamma^0 g$$

(9.45) "Infinitesimal Axial Transformation"

$$\psi \longrightarrow \psi' = (1 + i\epsilon \gamma^5) \psi$$

$$\bar{\psi} \longrightarrow \bar{\psi}' = \bar{\psi} (1 + i\epsilon \gamma^5)$$

(pg 548) "Transformation"

$$J = 1 + iM$$

(pg 548) "Matrix elements"

$$M_{nm} = \langle \phi_n, \epsilon \gamma^5 \phi_m \rangle$$

Verify $c' = J \cdot c$:

$$c' = J \cdot c$$

$$= (1 + iM) c$$

$$= (1 + iM_{nm}) c$$

$$= (1 + i \langle \phi_n, \epsilon \gamma^5 \phi_m \rangle) c$$

$$= (1 + i \int d^d x \phi_n^\dagger \gamma^0 \epsilon \gamma^5 \phi_m) c$$

$$= \left[\int d^d x \bar{\phi}_n (1 + i\epsilon \gamma^5) \gamma^0 \phi_m \right] c$$

Quantum Mechanics - Postulate #1:

"Wave functions satisfy a normal

Condition: $\int_{-\infty}^{\infty} \Psi^*(r, t) \cdot \Psi(r, t) d\tau = 1$

with single-values, continuous and finite wave functions"

Cite: Schrodinger, Erwin

"An Undulatory Theory of the
Mechanics of Atoms and
Molecules"

Physical Review (1926)

$$c' = \int d^3x \overline{\Phi_n} \Phi_m \cdot c$$

Exercise #6:
(pg 550)

(9.47) "Trace with Convergence
generating factor"

$$\text{tr}(e^{\gamma^5} e^{\delta \Box^2}) = \int \frac{d^d x d^d p}{(2\pi)^d} \text{tr}(e(x) \cdot \gamma^5 \cdot e^{-\delta(p^2 + i F_{\mu\nu}(x) \gamma^{\mu} \gamma^{\nu} / 2)})$$

@ d=4

$$\text{tr}(e \gamma^5) = \int \frac{d^4 x d^4 p}{(2\pi)^4} \text{tr}(e(x) \cdot \gamma^5)$$

$$= \int d^4 x e(x) F_{\mu\nu} \cdot F_{\rho\sigma} \epsilon^{\mu\nu\rho\sigma} \cdot \int \frac{d^4 p}{(2\pi)^4}$$

$$= \frac{1}{16\pi^2} \int d^4x \epsilon(x) \cdot F_{\mu\nu} \cdot 2\pi$$

(pg 779) "Dual Field Strength Tensor"

$$G = -*F$$

$$= -\frac{1}{4} \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} dx^\rho \wedge dx^\sigma$$

(9.42) "Change in axial current"

$$\partial_\mu \hat{J}_A^\mu = \frac{1}{2\pi} \epsilon^{\mu\nu} F_{\mu\nu}$$

(pg 546) "Two dimensional (Dual) Field Strength Tensor"

$$F = \frac{1}{2} F_{\mu\nu} \cdot dx^\mu \wedge dx^\nu$$

$$\text{tr}(\epsilon \gamma^5) = \frac{1}{8\pi} \int d^4x \epsilon(x) \cdot F_{\mu\nu}$$

$$= \frac{1}{32\pi} \int d^2x \epsilon(x) F_{\mu\nu} \cdot F_{\mu\nu} \epsilon^{\mu\nu\rho\sigma}$$

$$= \frac{1}{8\pi^2} \int \epsilon \cdot F \wedge F$$

Problem 9.4.1 - Parity, charge conjugation, and time-reversal symmetry:

a) (9.20) "Gamma Matrices for Weyl representations"

$$\gamma^0 = \tau_1, \quad \gamma^i = -i\tau_2 \otimes \sigma_i$$

(9.22) "Gamma Matrices for Majorana representations"

$$\gamma^0 = \sigma_2 \otimes \tau_1, \quad \gamma^1 = i\sigma_3, \quad \gamma^2 = \sigma_2 \otimes (-i\tau_2), \quad \gamma^3 = -i\sigma_1$$

where $\tau_1 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$

$$\sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$\tau_2 = \begin{bmatrix} 0 & 2i \\ 2i & 0 \end{bmatrix}$$

Weyl representation			
	Adjoint ^(†)	Transpose ^(T)	Complex ^(*)
γ^0	True [$\tau_1 = \tau_1^\dagger$]	True [$\tau_1 = \tau_1^T$]	True [$\tau_1 = \tau_1^*$]
γ^1	False	False	False
γ^2	False	True	False
γ^3	False	False	True

Majorana Representation			
	Adjoint ^(†)	Transpose ^(T)	Complex ^(*)
γ^0	True	False	False
γ^1	False	True	False
γ^2	False	True	False
γ^3	False	True	False

b) (9.58) "Dirac action"

$$S[\psi] = \int d^4x \bar{\psi} (i \not{D} - e \not{A} - m) \psi$$

where $A =$ "vector potential"

$$= A_\mu \gamma^\mu$$

$\not{D} =$ "Dirac operator"

$$= (\not{\partial} - i \not{A}) \gamma^\mu$$

$\bar{\psi} =$ "Dirac conjugate spinor"

$$= \psi^\dagger \gamma^0$$

(From problem) "Transformation"

$$\psi_x^\dagger = \bar{U}_{xj} \psi_j^\dagger$$

$$\psi_{xj} = U_{xi} \psi_i$$

(from problem) "Arbitrary assignment"

$$U_P = U_P(\gamma^\mu)$$

(9.59) "Formal Symmetry"

$$\psi(x) \rightarrow \psi'_x(x) \equiv X \cdot \psi(\Lambda_x^{-1} x) X^{-1}$$

$$\begin{aligned} S[\psi] &= \int d^4x \bar{\psi} (i \not{D} - e \not{A} - m) \psi \\ &= \int d^4x \psi^\dagger{}^T \gamma^0{}^T (i \not{D} - e \not{A} - m) \psi \\ &= \int d^4x \bar{U} \psi \gamma^0{}^T (i \not{D} - e \not{A} - m) \psi \\ &= \int d^4x \bar{U} \psi (\Lambda^{-1} x) (\gamma^0 (i \not{D} - e \not{A} - m) U \psi (\Lambda^{-1} x) \gamma^0 \\ &= \int d^4x \bar{U} \psi(x) \gamma^0 (i \partial_\mu \Lambda^\mu_\nu \gamma^\nu - m) U \psi(x) \gamma^0 \\ &= \int d^4x \bar{\psi} (\gamma_0 U^\dagger \gamma^0) (i \partial_\mu (\Lambda \gamma)^\mu - m) U \psi \end{aligned}$$

If $(\Lambda \gamma)^0 = \gamma^0$ and $(\Lambda \gamma)^i = -\gamma^i$

then $[U, \gamma^i] = 0$

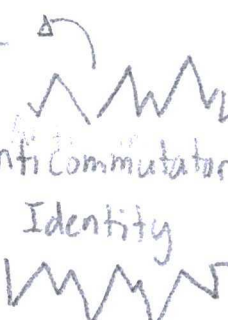
c) Charge conjugation:

(from problem) "Formal Transformation"

$$\psi_c^\dagger = U_{c, \gamma} \psi^\dagger$$

$$\begin{aligned} S[\psi_c] &= \int d^4x \bar{\psi} (i \not{D} - e \not{A} - m) \psi \\ &= \int d^4x U^\dagger \gamma^0 (i \partial_\mu \gamma^\mu - m) U \psi^\dagger @ A=0 \end{aligned}$$

$$= - \int \psi^{\dagger T} U^T (-i \partial_\mu \gamma^\mu T - m) \gamma^0 T U \psi$$



 $\sum$ Anticommutator Identity

The transformations in Table 9.1 relate through a Hermitian and Transpose, but only if $\gamma^2 = 0$. Majorana representation requires γ^0 and γ^i near zero, since not unitary between Hermitian and Transpose.

d) Time-reversal:

$$\begin{aligned}
 S[\psi_T] &= \int d^4x \bar{\psi}_T (-i \partial_\mu \gamma^\mu - m) \psi_T \\
 &= \int d^4x \bar{\psi} U^T (\Lambda_x^{-1}) \gamma^0 (-i \partial_\mu (\Lambda \bar{\gamma})^\mu - m) U \psi (\Lambda_x^{-1}) \gamma^0
 \end{aligned}$$

By the table, $(\Lambda \gamma)^{\dagger, 3} = \gamma^{\dagger, 3} = 0$
 and $(\Lambda \gamma)^{\dagger, 2} = -\gamma^{\dagger, 2} = 0$

U acts on spin by commutation

$$\begin{aligned}
 \text{With } [U, \gamma^{\dagger, 3}] &= U \gamma^{\dagger, 3} - \gamma^{\dagger, 3} U \\
 &= \gamma^{\dagger, 3} \gamma^{\dagger, 3} - \gamma^{\dagger, 3} \gamma^{\dagger, 3} \\
 &= -i \sigma_2 + i \sigma_2
 \end{aligned}$$

$$= 0$$

e) Vector Potential coupling:

— If $A_\mu \rightarrow A(\Lambda P)_\mu^\nu$, the vector potential

$\nabla \times A = B$ changes to no sign. But

by invariance E becomes $-E$. Likewise,

eA follows same output.

Problem 9.4.2: Quantization of Dirac Action:

a) Momentum:

(from problem) "Two dimensional Dirac Action"

$$S[\psi] = \int d^2x \bar{\psi} (i \partial_\mu \gamma^\mu - m) \psi$$

$$\text{where } \gamma^0 = \sigma_1$$

$$\gamma^1 = i\sigma_2$$

$$= \int d^2x L(\psi, \dot{\psi})$$

(from problem) "Momentum"

<u>Lagrangian:</u> $\frac{\partial L}{\partial \dot{q}} = \frac{\partial L}{\partial \dot{q}}$ $\uparrow$ $\uparrow$ Momentum Force

$$\pi = p = \frac{\partial L}{\partial \dot{\psi}}$$

$$= \frac{\partial \bar{\psi}}{\partial \dot{\psi}} (i \partial_\mu \gamma^\mu - m) \psi$$

$$= i\dot{\psi}$$

Hamiltonian Density:

(1.10) "Hamiltonian"

$$\begin{aligned} H(\psi, \pi) &= p\dot{q} - L(p, q) \\ &= \pi \dot{\psi} - L(\pi, \psi) \end{aligned}$$

Hamiltonian Action:

$$S[\psi] = \int d^2x \pi \dot{\psi} - H(\pi, \psi)$$

@ $d = 4$, The electronic system has more states with half-integer spin for electrons.

b) (from problem) "Bosonic relation"

$$[\psi_i(x), \psi_j^\dagger(y)] = \delta_{ij} \delta(x-y)$$

(from problem) "Fermionic relation"

$$[\psi_i(x), \psi_j^\dagger(y)] = \delta_{ij} \delta(x-y)$$

The action, $S[\psi] = \int d^3x (\pi \dot{\psi} - H(\pi, \psi))$ has no convergence with infinite mass.

An action, $S[\psi, \pi] = \int d^3x (\pi \dot{\psi} - \psi^\dagger (i\partial_x \sigma_3 + m\sigma_1) \psi)$

never converges.

Helmholtz Free Energy:

$$F = kT \ln Z$$
$$= kT \ln \int d\psi e^{-S[\psi, \pi]}$$

The Free energy is indefinite from infinite action. Other actions match a boson with additional properties, mass, spin, and charge.

Problem 9.4.3- Chiral anomaly and Topological Quantization:

(9.39) "1+1 Dirac Action Integral"

$$S[\psi] = \int d^2x \psi^\dagger ((i\partial_t + A_0) + \sigma_3(i\partial_1 + A_1) - m\sigma_1) \psi$$

(from problem) "Parameters"

L = "Spatial circle with radius"

Φ_0 = "Flux"

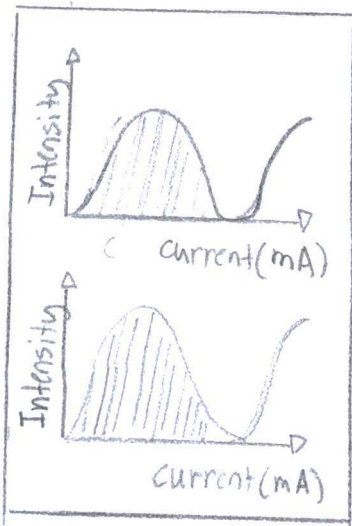
$$= h/e$$

$$= 2\pi$$

a) $U(1)$ fiber bundle fits the action because $U(1) \times U(1)$ similarities to a

circular torus fiber. The electric field, $E = [2tA]$ varies by potential, specifically vector potential $[A = \Phi_0/L]$. Section 10.5.2 suggests a gauge transformation in the action, $[\psi \rightarrow e^{i\Phi(t)\frac{x}{L}} \cdot \psi]$. The shift in potential by 2π or $t + 2\pi$ is necessary for invariant values. The magnetic vector potential $[B = \nabla \times A]$ is also invariant with exact transformation in 2π or $2\pi/L$.

b) (9.54) "Chern Class"



$$\int_m dS_A = 2 \int \text{ch}_{d/2}(F) = 2 \cdot \text{ch}_{d/2}(F)$$

"Area under curve relates to electron charge."

Also, charge appears by decomposition of current, $j(x,t) = j(x) + j(t)$

"An integral state with countable values, Chern numbers. The values describe topology in sample"

Integer values
"Chern Class"
 $\int \text{current} = \text{integer} \cdot \text{Area}$
Gauge Theory
 $j_2 = j_1 + n \cdot \Delta j$

c) Weyl Modes:

(from problem) "Weyl Equation"

$$[-\partial_\tau + c(-i\partial_x - \phi(\tau)/L)]\psi_c(x,t) = 0$$

() "Dirac operator"

$$iD = (i\partial_\mu + A_\mu)\gamma^\mu$$

$$= -\partial_\tau \psi_c(x,t) - i(\partial_x - iA_x(\tau)) \psi_c(x,t)$$

$$\text{when } A_x = \phi(\tau)/L$$

Eigen values:

$$-i(\partial_x - iA_x(\tau)) \psi_c(x,t) = \lambda \psi_c(x,t)$$

$$\lambda = (\partial_x - iA_x(\tau)), 0$$

$$\psi_c(x,t) = e^{ikx}$$

$$\text{if } k = \frac{2\pi n}{L}$$

Zero Mode:

At $\lambda=0$, the n-value is zero and the potential too.

$$(-\partial_\tau + c(-i\partial_x - \frac{\phi(\tau)}{L}))\psi_c(x,t)$$

$$= (-\partial_\tau + c(-i\partial_x - \frac{\phi(\tau)}{L}))e^{ikx/L}$$

$$= (-\partial_\tau + c(-i\partial_x - \frac{\phi(\tau)}{L})) e^{\frac{i2\pi n \cdot 0}{L}}$$

$$= 0$$

Problem 9.4.4 - Parity Anomaly:

(from problem) "one-dimensional Dirac action"

$$\int dx \bar{\psi} (-i\partial_x - A) \psi$$

(from problem) "Gauge Transformation"

$$A = \phi(\tau)/L$$

$$= 2\pi W/L \quad \text{with } \int dx A = 2\pi W$$

(from problem) "Chern Number"

$$\int dx A = 2\pi W \quad \text{where } W = \text{Chern \#}$$

$$= 1$$

a) Weyl Modes:

$$(-i\partial_x - A(\tau)) \psi(\tau) = \lambda \psi(\tau)$$

$$\lambda = i\partial_x - A \text{ or } 0$$

$$\psi = e^{ikx/L}$$

The eigenvalue has multiple values (positive/negative) at $A(\tau) = 0$, including

real or complex.

b) (from problem) "Chern-Simons Term"

$$D \equiv \partial_x - iA$$

(from problem) "Regularized Chern Term"

$$\bar{\Phi}(-iD + iM)\Phi$$

UV-singularities (Eigenstates):

$$\begin{aligned}\langle iD + iM \rangle &= \int dx \bar{\Phi}^\dagger (iD + iM)\Phi \\ &= \langle \lambda + iM \rangle\end{aligned}$$

The expected value is an average, such as $\prod_{i=1}^n \frac{\lambda_i}{\lambda + M}$.

A large wavelength ($\lambda \gg M$) is not possible from the average or expectation. While small masses approximate small wavelengths, but conceptually units are far from correct.

c) (from part c) "Abelian Chern-Simons"

$$S_{CS}[A] = 2\pi k \int \text{ch}_{(d+1)/2}(F)$$

where (10.49) "Chern numbers"

$$\text{ch}_1(F) = \frac{i}{2\pi} \text{tr}(F) = \frac{i}{2\pi} d \text{tr}(A)$$

$$\text{ch}_2(F) = -\frac{1}{8\pi^2} \text{tr}(F \wedge F)$$

$$= -\frac{1}{8\pi^2} d \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right)$$

knowing $k = \frac{2\pi X}{L}$

(pg 194) "Partition Function"

$$Z = \int dA e^{-S[A]}$$

$$= \int dA e^{-2\pi k \int \text{ch}_{(d+1)/2}(F)}$$

$$= \int dA e^{-2\pi k \int \frac{i}{2\pi} d \text{tr}(A)} \quad @d=1$$

Sign changes occur from a large gauge transformation. The wavevector ($k = 2\pi/\lambda$) is negative in opposite direction, possibly with a current bias.

Expansion of A:

$$\begin{aligned} iS[\Phi] &= i \int dx \bar{\Phi} [-iD + iM] \Phi \\ &= i \int dx \bar{\Phi} \text{tr} \ln(p - A + iM) \Phi \\ &= -i \int dx \bar{\Phi} \text{tr} (A(p + iM)^{-1}) \quad \dots p \ll 1 \text{ and } M \ll 1 \\ &= -i \int \frac{dx dp}{2\pi} A(x) \frac{1}{p + iM} + \dots \\ &\cong - \int dx dp \text{ch}_1(x) \frac{1}{p + iM} \end{aligned}$$

Problem 9.4.5 - Functional Bosonization:

(3.86) "One-dimensional electron gas action"

$$\begin{aligned} S_0[\Psi] &= \int d^2x \Psi^\dagger (\sigma_0 \partial_{x_0} - i\sigma_3 \partial_{x_1}) \Psi \\ &= \int d^2x \bar{\Psi} (\sigma_1 \partial_{x_0} - \sigma_2 \partial_{x_1}) \Psi \end{aligned}$$

(3.90) "Interaction Contribution"

$$S_{\text{int}}[\Psi] = \frac{1}{2} \sum_j \int dx d\tau (g_2 \hat{p}_s \hat{p}_s + g_4 \hat{p}_s \hat{p}_s)$$

$$\text{where } p_s = \Psi_s^\dagger \Psi_s$$

(from description) "Source Term"

$$S_{\text{source}}[\Psi, j] \equiv \int \sum_{s=\pm} (\Psi_s^\dagger j_s + j_s^\dagger \Psi_s)$$

(from problem) "Matrix Representation"

$$S_{int}[\psi] = \frac{1}{2} \int d\tau dx (\hat{\psi}_+ \hat{\psi}_-) g \begin{pmatrix} \hat{\psi}_+ \\ \hat{\psi}_- \end{pmatrix}$$

$$g = \begin{pmatrix} g_4 & g_2 \\ g_2 & g_4 \end{pmatrix}$$

a) Fermionic Interaction:

$$S[\psi] = S_{int}[\psi] + S_0[\psi]$$

$$= \frac{1}{2} \int d\tau dx (\hat{\psi}_+ \hat{\psi}_-) g \begin{pmatrix} \hat{\psi}_+ \\ \hat{\psi}_- \end{pmatrix} + \int d^2x \bar{\psi} (\gamma^0 \partial_0 - \gamma^1 \partial_1) \psi$$

$$+ \int d^2x \bar{\psi} (\gamma^1 \partial_0 - \gamma^2 \partial_1) \psi$$

$$= \frac{1}{2} \int d\tau dx (\hat{\psi}_+ g_4 \hat{\psi}_+ + \hat{\psi}_- g_2 \hat{\psi}_+ + \hat{\psi}_+ g_2 \hat{\psi}_- + \hat{\psi}_- g_4 \hat{\psi}_-) + \int d^2x \bar{\psi} (\gamma^1 \partial_0 - \gamma^2 \partial_1) \psi$$

$$+ \int d^2x \bar{\psi} (\gamma^1 \partial_0 - \gamma^2 \partial_1) \psi$$

$$= \frac{1}{2} \int d^2x \phi^T g^{-1} \phi + \int d^2x \bar{\psi} (i \not{D} - \phi) \psi$$

$$\dots g = g^T$$

$$\phi = \partial_\mu \gamma^\mu$$

$$\phi = -(2\xi - i2\eta)$$

b) (9.60) "Effective Action"

$$S[\psi, \phi] = \frac{1}{2} \int d^2x \phi^\dagger g^{-1} \phi + \int d^2x \bar{\psi} (i \not{\partial} - \phi) \psi$$

$$= \frac{1}{2} \int d^2x \phi^\dagger g^{-1} \phi - \text{tr} \ln(\not{\partial} - i \phi)$$

$$= \frac{1}{2} \int d^2x \phi^\dagger g^{-1} \phi - \frac{1}{2} \text{tr} \ln(\not{\partial} \phi \not{\partial} \phi) + \dots$$

+ ln

<u>Logarithm</u> <u>Expansion</u> $\ln(1-x) = -x - \frac{x^2}{2}$
--

$$= \frac{1}{2} \sum_k \phi_k (g^{-1} + \delta \Pi_{s,k}) \phi_k$$

when $\Pi_k = \left\{ \frac{\delta_{ss'}}{2\pi} \frac{q}{-is\omega + q} \right\}$

c) (9.63) "Green Function"

$$\hat{G}_{ss}(X, \hat{X}) = (2\pi a)^{-1} \left\langle e^{-i\langle \phi + s\theta \rangle(X)} e^{i(\phi + s\theta)(X')} \right\rangle$$

(9.63) "Green Function generalized"

$$\hat{G}_{ss}(X, \hat{X}) = \left\langle \prod_{\text{mod}} e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right\rangle$$

$$Z = \iint D\Xi D\Gamma e^{-S[\Xi] - S[\Gamma]} \hat{G}_{ss}(X, X')$$

$$= \iint D\Xi D\Gamma e^{-S[\Xi] - S[\Gamma]} e^{-\int d^2x A \dots}$$

$$A = \left\langle \prod e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right\rangle$$

For shift, $\phi \rightarrow \phi - \xi$

$$\theta \rightarrow \theta - \eta$$

$$\Xi \rightarrow \Xi - \Gamma$$

$$Z = \iint D\Xi D\Gamma e^{-S_0[\Xi - \Gamma] - S[\Gamma] - S[\Xi]} \int d^2x \left(j e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} + e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right)$$

(9.64) "Bosonic Doublet"

$$S_0[\Xi] = \frac{1}{2} \sum_k \Xi_k^T K_k \Xi_{-k} \quad K_k = \frac{1}{\pi} (q^2 - i q \omega \sigma_1)$$

$$Z = \iint D\Xi D\Gamma e^{-(\frac{1}{2} \sum_k \Xi_k^T K_k \Xi_{-k} + \Gamma_k^T (D_k^T g^{-1} D_k) \Gamma_{-k} + \Xi_k^T K_k \Gamma_{-k})}$$

$$+ \Gamma_k^T K_k \Xi_{-k} \int d^2x \left(j e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} + e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right)$$

$$= \int D\Xi e^{-\frac{1}{2} \sum_k \Xi_k^T [K_k + 2q^2(g_4 - g_2\sigma_3)] \Xi_{-k}} \int d^2x \left(j e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} + e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right)$$

$$= \int D(\phi, \theta) e^{-\frac{1}{2\pi} \sum_k \left(\frac{q^2[1 + 2\pi(g_4 - g_2)] - i q \omega}{-i q \omega} \frac{q^2[1 + 2\pi(g_4 + g_2)]}{q^2[1 + 2\pi(g_4 + g_2)]} \right) \begin{pmatrix} \phi \\ \theta \end{pmatrix}} \int d^2x \left(j e^{-i(\xi + \phi + (\eta + \theta)\sigma_3)} + e^{i(\xi + \phi + (\eta + \theta)\sigma_3)} \right)$$

Note: Best stuff from working through Chapter 9, include Chern numbers, Majorana and Weyl wavefunctions.